FinMatrix

AI-Powered Stock Analysis Platform

AI Agent Integration Planning Document

Homework Assignment — Foundations Module

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GitHub: github.com/alpbel0/FinMatrix

1. Project Overview

1.1 Website Topic and Purpose

FinMatrix is an AI-powered financial analysis platform designed for Turkish stock market (BIST) investors. The platform allows users to interact with financial data using natural language queries, receiving dynamically generated charts and analyses in response. Rather than passively browsing static data, users can ask questions like "Compare the net profit of THYAO and ASELS over the last 3 years" and receive instant visual analysis powered by real financial data.

The platform presents data-driven, unbiased analysis and lets the user make their own investment decisions. FinMatrix never makes buy or sell recommendations.

1.2 Target Users

- **Individual investors (beginner to advanced)** who want to analyze BIST-listed stocks
- **Finance students and researchers** seeking structured financial data
- **Anyone who wants to understand** a company's financial health through its KAP filings

1.3 Core Features

Feature	Description
News Feed	Real-time KAP disclosure notifications, filtered by watchlist
Stock Dashboard	Price charts, candlestick graphs, and financial metric cards per stock
AI Chat Interface	Natural language Q&A with inline dynamically generated charts
Watchlist Management	Add/remove stocks, toggle Telegram notification per stock
Telegram Bot	Automated KAP filing alerts delivered to user's Telegram

2. Website Screenshots

2.1 News Feed Page

The main feed displays real-time KAP disclosures and financial news for stocks in the user's watchlist. The left sidebar shows tracked stocks with live prices, while the right panel shows BIST100 index data and market movers.

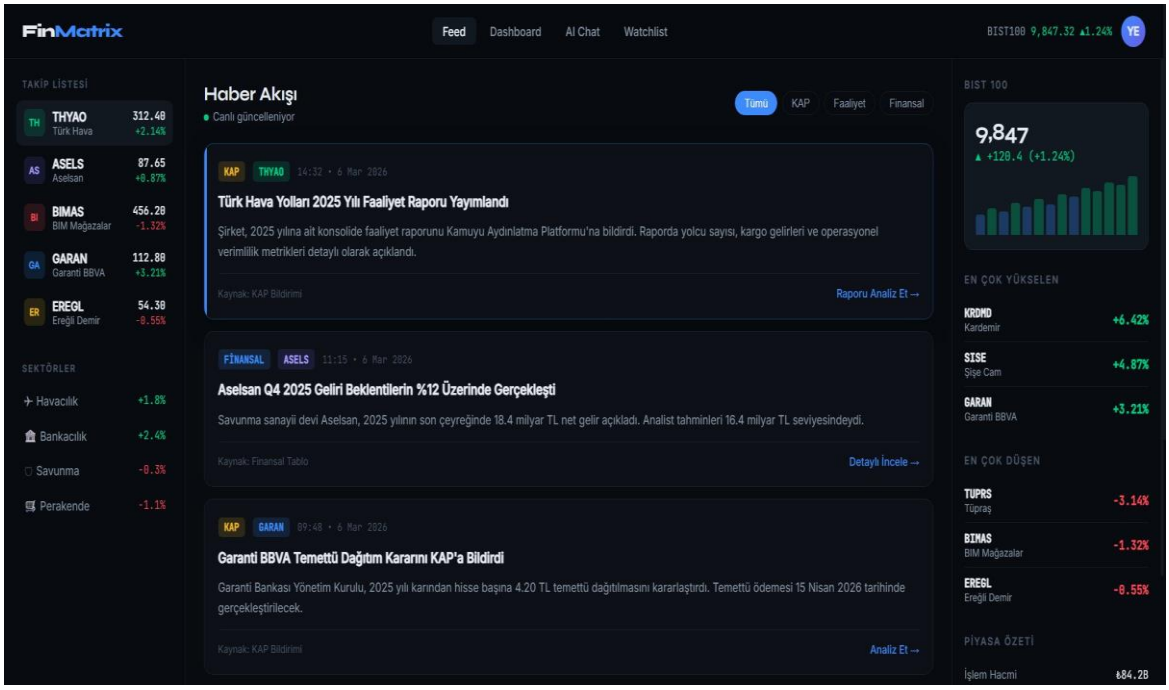

Figure 1 — News Feed Page: KAP disclosures, watchlist sidebar, BIST100 panel

2.2 Stock Dashboard

The dashboard provides a deep-dive view for any selected stock. Users can view price history across multiple timeframes, financial metric cards (P/E, ROE, Net Profit), quarterly profit bar charts, and a full financial summary table.

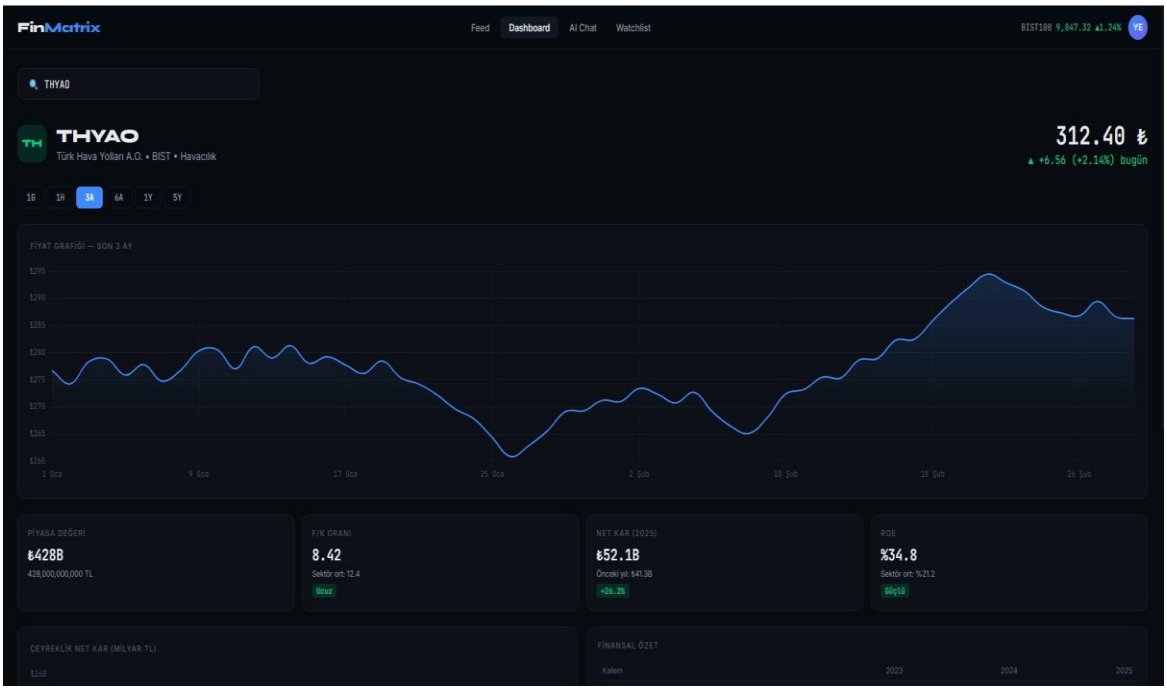

Figure 2 — Dashboard Page: THYAO price chart, financial metrics, quarterly profit chart

2.3 AI Chat Interface

The chat interface is the core of FinMatrix. Users type natural language questions and receive responses that include inline dynamically generated charts — similar to Claude's artifact system. The right panel shows the KAP source documents used to generate each response, ensuring full transparency.

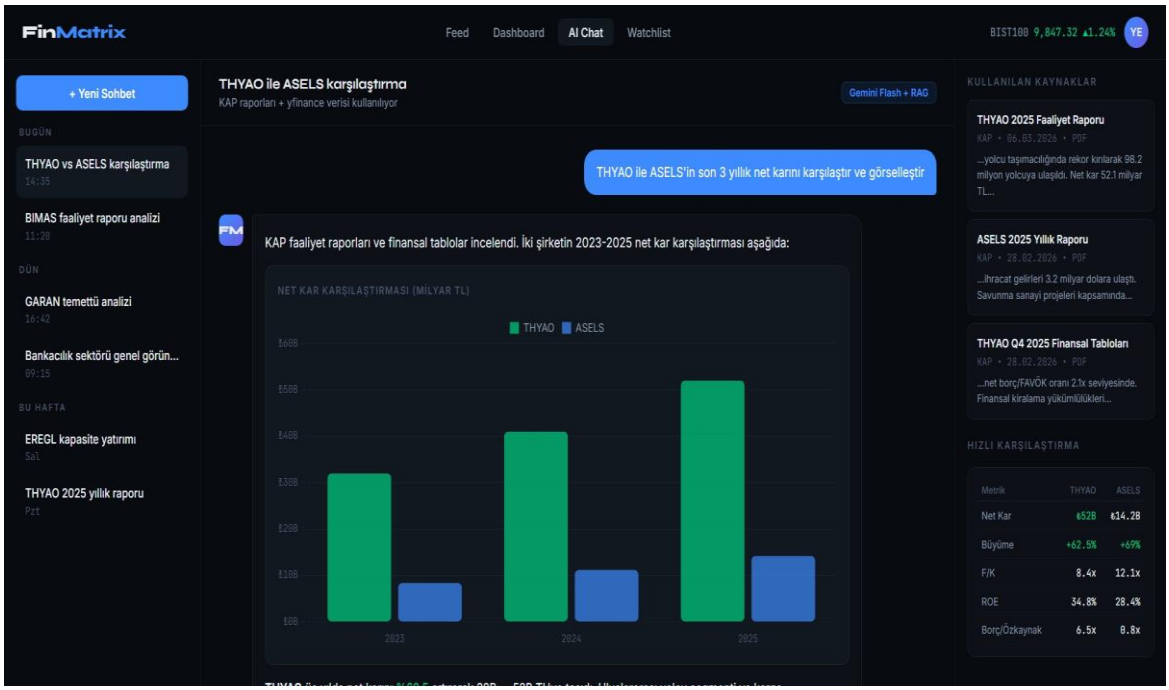

Figure 3 — AI Chat Page: THYAO vs ASELS net profit comparison with inline bar chart

2.4 Watchlist Page

The watchlist page gives users a bird's-eye view of all tracked stocks. Each card shows live price, sparkline chart, key metrics (P/E, market cap, volume), and a Telegram notification toggle. Users can manage their entire watchlist from this single page.

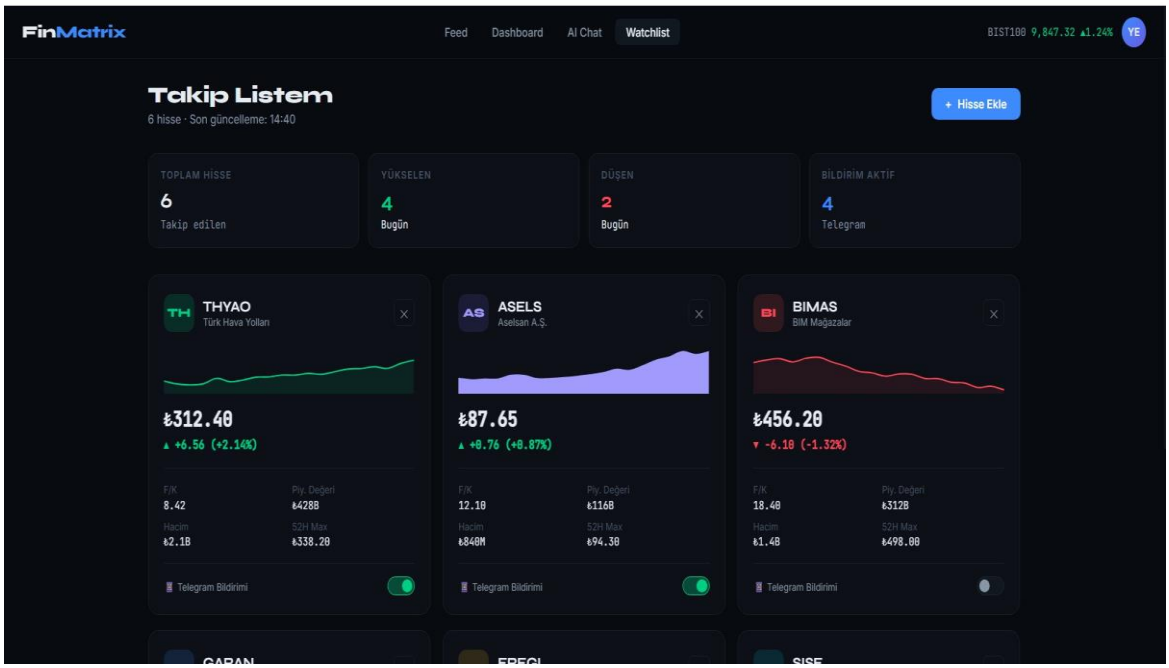

Figure 4 — Watchlist Page: Stock cards with sparklines and Telegram notification toggles

3. AI Agent Concept

3.1 Problem Statement

Turkish retail investors face a critical information problem: KAP publishes hundreds of filings every day in dense PDF format — annual reports, financial statements, special disclosures — and nobody has time to read them all. Meanwhile, yfinance provides raw numerical data but no interpretation.

The FinMatrix AI agent solves this by:

- **Automatically ingesting** and indexing KAP PDF filings into a vector database
- **Allowing users** to query this knowledge base in natural language
- **Generating visual comparisons** and analyses on demand
- **Detecting and blocking** hallucinated responses before they reach the user

3.2 Multi-Agent Architecture with Role Separation

FinMatrix uses a strict role-separation architecture to prevent agent chaos and API cost inflation. Each framework handles only what it is best at:

Framework	Role	Responsibility
LangGraph	Orchestrator	Receives user query, routes to correct agent, merges results, triggers Judge
CrewAI	Text Analyst	Reads KAP PDFs from ChromaDB, summarizes, extracts risk/vision insights
AutoGen	Code Executor	Writes and runs Python for numerical analysis and chart generation
Judge Agent	Evaluator	Scores response against source documents, blocks hallucinations

3.3 User Interaction Modes

- **Chat Interface:** Natural language questions → AI response with inline charts
- **Inline Chart Generation:** AutoGen writes Python to generate charts, rendered inside chat
- **Background Automation:** Data pipeline runs silently — always pre-indexed, always ready
- **Telegram Notifications:** KAP filing alerts delivered automatically, managed from Watchlist

4. System Architecture

4.1 Two-Pipeline Design

Pipeline 1 — Background Data Ingestion (Scheduler-driven):

- **Scheduler triggers periodically (nightly)**
- **KAP scraper fetches new PDF filings for tracked stocks**
- **yfinance API fetches updated financial tables and price history**
- **PDFs are parsed and chunked, embedded into ChromaDB (each stock = separate collection)**
- **SQL database tracks sync status per document: PENDING → SUCCESS / FAILED**

Pipeline 2 — Real-Time Query Processing (User-triggered):

- **User submits natural language query via chat UI**
- **LangGraph Orchestrator analyzes query intent and routes to agents**
- **MCP data layer performs semantic search — only relevant chunks retrieved**
- **CrewAI handles text analysis; AutoGen handles charts and calculations**
- **Judge Agent scores response for hallucination (BERTScore / LLM-as-Judge)**
- **Verified response delivered to user; failed responses trigger automatic retry**

4.2 Tech Stack

Component	Technology
Frontend	React + Chart.js
Backend API	FastAPI (Python)
Orchestration	LangGraph
Text Analysis	CrewAI
Code Execution / Charts	AutoGen
Vector Database	ChromaDB
Data Layer	Model Context Protocol (MCP)
Relational Database	PostgreSQL
EvalOps / Judge	LLM-as-Judge + BERTScore + ROUGE
Stock Data	yfinance
Financial Filings	KAP Web Scraping
LLM	Gemini Flash
Notifications	Telegram Bot API
Hosting	Digital Ocean

4.3 EvalOps Layer

The Judge Agent sits between the agent pipeline and the user. It evaluates every response against source KAP documents before delivery:

- **BERTScore:** semantic similarity between response and source chunks
- **ROUGE Score:** n-gram overlap for factual consistency
- **LLM-as-Judge:** secondary model evaluates plausibility and accuracy
- **is_hallucinated flag:** Boolean stored in Eval_Logs for monitoring and auditing